

6.2

Thème 1. Soient $f: V \rightarrow V$ telle $V = \text{span}\{v_1, v_2\}$, $v_1 = (2, 1, 0, 1)^T$, $v_2 = (1, 0, 1, 2)^T$
 et telle $f(v_1) = (4, 1, 2, 5)^T$, $f(v_2) = (2, 2, -2, -2)^T$ et on cherche $f((5, 1, 3, 7)^T)$

On écrit $f(v_1)$ et $f(v_2) \in V$ donc on a, b tels que $f(v_1) = av_1 + bv_2$
 $(4, 1, 2, 5)^T = a(2, 1, 0, 1)^T + b(1, 0, 1, 2)^T = (2a+b, a+0b, 0a+b, a+2b)^T = (2a+b, a, b, a+2b)^T$

$$4 = 2a + b, 1 = a, 2 = b, 5 = a + 2b$$

$$2 \cdot 1 + 2 = 2 + 2 = 4$$

$$1 + 2 \cdot 2 = 1 + 4 = 5 \quad \text{Donc } f(v_1) = 1v_1 + 2v_2$$

et donc on a c, d tels que $f(v_2) = cv_1 + dv_2$ mais on a $f(v_2) = (2, 2, -2, -2)^T$

$$(2, 2, -2, -2)^T = c(2, 1, 0, 1)^T + d(1, 0, 1, 2)^T = (2c+d, c, 0c+d, c+2d)^T$$

$$2 = 2c + d, 2 = c, -2 = d, -2 = c + 2d \quad (c=2, d=-2)$$

$$2 \cdot 2 + (-2) = 4 - 2 = 2$$

$$2 + 2 \cdot (-2) = 2 - 4 = -2 \quad \text{Donc } f(v_2) = 2v_1 + (-2) \cdot v_2 = 2v_1 - 2v_2$$

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

Donc on a $\vec{v} = A\vec{v}$ et on cherche B

$$\text{soient } u = p \cdot v_1 + q \cdot v_2$$

$$1 = 2p + q = 2 + q \Rightarrow q = -1$$

$$1 = p \Rightarrow p = 1$$

$$3 = 2p + q = 2 + (-1) = 1$$

$$1 = 1p + 2q = 1 + 0 = 1 \quad \text{Donc } [u]_B = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ que } [f(u)]_B = A \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\det(A) = -6 \quad \chi^2 - \lambda - 6 = 0 \quad \lambda_1 = 2, \lambda_2 = -3$$

$$P_{\lambda=2} = 2(2, 1)^T$$

$$P_{\lambda=-3} = -3(1, -2)^T$$

$$(1, -1)^T = 1 \cdot (2, 1)^T + (-1) \cdot (1, -2)^T$$

$$[f(u)]_B = 1 \cdot 2 \cdot (2, 1)^T + (-1) \cdot (-3) \cdot (1, -2)^T = \begin{pmatrix} 2 \cdot 2 - (-3) \\ 1 \cdot 2 + 2 \cdot (-3) \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$a_7 \quad f(u) = (2^{n+1} - (-3)^n)u_1 + (2^n + 2(-3)^n)u_2$$

$$2(2^{n+1} - (-3)^n) + (2^n + 2(-3)^n) = 2^{n+2} + 2^n = 4 \cdot 2^n$$

$$1(2^{n+1} - (-3)^n) + 0 = 2^{n+1} - (-3)^n$$

$$0 + 1(2^n + 2(-3)^n) = 2^n + 2(-3)^n$$

$$-1(2^{n+1} - (-3)^n) + 2(2^n + 2(-3)^n) = 2^{n+1} - (-3)^n + 2^{n+1} + 4(-3)^n = 2^{n+2} + 3(-3)^n$$

$$f^m((1, 1, 1)^T) = \begin{pmatrix} 5 \cdot 2^m \\ 2^{m+1} + 3^m \\ 2^m - 6^m \\ 2^{n+2} - 4^m \end{pmatrix}$$